

Exp: 4- Write the Python Program for Cript-Arithmetic Problem.

Aim:

To write a python program for Cript-Arithmetic problem.

Algorithm:

Step 1: Write down the arithmetic problem in full, including any carry-overs. **Step 2:** Replace each letter with a unique digit from 0 to 9.

Step 3: Generate all possible combinations of digits for each letter, subject to any constraints ' from the carry-overs and the rules of arithmetic.

Step 4: For each combination of digits, evaluate the arithmetic problem and check if it satisfies the given constraints.

Step 5: If a solution is found, output the values of the letters and the arithmetic problem, and exit the algorithm.

Step 6: If no solution is found, output a message indicating that no solution exists.

Program:

```
from itertools import permutations

letters = "SENDMORY"

for p in permutations("0123456789", 8):

    s, e, n, d, m, o, r, y = map(int, p)

    # Leading digit cannot be zero
    if s == 0 or m == 0:
        continue

    SEND = 1000*s + 100*e + 10*n + d
    MORE = 1000*m + 100*o + 10*r + e
```

$$\text{MONEY} = 10000*m + 1000*o + 100*n + 10*e + y$$

if SEND + MORE == MONEY:

```
print("Solution Found")
print("SEND =", SEND)
print("MORE =", MORE)
print("MONEY =", MONEY)
```

```
print("\nLetter Mapping")
print("S =", s)
print("E =", e)
print("N =", n)
print("D =", d)
print("M =", m)
print("O =", o)
print("R =", r)
print("Y =", y)
```

break

Output:

```
Solution Found
SEND = 9567
MORE = 1085
MONEY = 10652

Letter Mapping
S = 9
E = 5
N = 6
D = 7
M = 1
O = 0
R = 8
Y = 2
```

After defining the `solve_cryptarithmic` function, the code provides an example usage using a cryptarithmic puzzle: "SEND + MORE = MONEY." It calls the `solve_cryptarithmic` function with the puzzle string and stores the result in the mapping variable.

If a solution is found (i.e., mapping is not None), it prints the mapping dictionary and the puzzle string with the letters replaced by their corresponding digits using the `translate` method.

If no solution is found, it prints "No solution found."

Result:

The Cryptarithmic puzzle was solved successfully using AI search. A valid assignment of digits to letters was obtained, satisfying the equation **SEND + MORE = MONEY**